

```
-creating the database ----
```

```
MariaDB [(none)]> CREATE DATABASE Projects ;
```

Query OK, 1 row affected (0.007 sec)

```
MariaDB [(none)]> USE Projects;
```

Database changed

```
-----CREATING TABLES-----
```

```
CREATE TABLE Employee (
```

empNo VARCHAR(10) NOT NULL,

fName VARCHAR(50) NOT NULL,

```
lName    VARCHAR(50)    NOT NULL,
```

```
address VARCHAR(100),
```

DOB DATE,

```
sex      CHAR(1)      NOT NULL,
```

```
position VARCHAR(20) NOT NULL,
```

deptNo VARCHAR(10),

```
CONSTRAINT pk_employee PRIMARY KEY (empNo),
```

```
CONSTRAINT chk_sex      CHECK (sex IN ('M', 'F')),
```

```
CONSTRAINT chk_position CHECK (position IN (
    'Manager',
    'Team Leader',
    'Analyst',
    'Software Developer'))
```

);

Query OK, 0 rows affected, 1 warning (0.063 sec)

QUERY:

```
CREATE TABLE Department (
```

deptNo VARCHAR(10) NOT NULL,

deptName VARCHAR(50) NOT NULL,

```
mgrEmpNo VARCHAR(10),
```

```
CONSTRAINT pk_department PRIMARY KEY (deptNo),
```

```
CONSTRAINT fk_dept_mgr FOREIGN KEY (mgrEmpNo)
REFERENCES Employee(empNo)
```

$$);$$

OUTPUT:

Query OK, 0 rows affected, 1 warning (0.039 sec)

Query :

MariaDB [Projects]> ALTER TABLE Employee

-> ADD CONSTRAINT fk_emp_dept

-> FOREIGN KEY (deptNo) REFERENCES Department(deptNo);

output:

Query OK, 0 rows affected (0.075 sec)

Records: 0 Duplicates: 0 Warnings: 0

QUERY:

CREATE TABLE WorksOn (

empNo VARCHAR(10) NOT NULL,

projNo VARCHAR(10) NOT NULL,

dateWorked DATE NOT NULL,

hoursWorked INT NOT NULL,

CONSTRAINT pk_workson PRIMARY KEY (empNo, projNo, dateWorked),

CONSTRAINT fk_wo_emp FOREIGN KEY (empNo)

REFERENCES Employee(empNo),

CONSTRAINT fk_wo_proj FOREIGN KEY (projNo)

REFERENCES Project(projNo),

CONSTRAINT chk_hours CHECK (hoursWorked BETWEEN 0 AND 40)

);

=====

VERIFY TABLES WERE CREATED

=====

QUERY:

SHOW TABLES;

OUTPUT:

+-----+

| Tables_in_projects |

+-----+

| department |

```

| employee      |
| project      |
| workson      |
+-----+
4 rows in set (0.027 sec)

```

```

-----
DESCRIBE EACH TABLE
-----

```

QUERY:

DESCRIBE Employee;

OUTPUT:

Field	Type	Null	Key	Default	Extra
empNo	varchar(10)	NO	PRI	NULL	
fName	varchar(50)	NO		NULL	
lName	varchar(50)	NO		NULL	
address	varchar(100)	YES		NULL	
DOB	date	YES		NULL	
sex	char(1)	NO		NULL	
position	varchar(20)	NO		NULL	
deptNo	varchar(10)	YES	MUL	NULL	

8 rows in set (0.00 sec)

QUERY:

DESCRIBE Department;

OUTPUT:

Field	Type	Null	Key	Default	Extra
deptNo	varchar(10)	NO	PRI	NULL	
deptName	varchar(50)	NO		NULL	
mgrEmpNo	varchar(10)	YES	MUL	NULL	

3 rows in set (0.00 sec)

QUERY:

DESCRIBE Project;

OUTPUT:

```
-----
+-----+-----+-----+-----+-----+-----+
| Field   | Type           | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| projNo  | varchar(10)    | NO   | PRI | NULL    |       |
| projName| varchar(100)   | NO   |     | NULL    |       |
| deptNo  | varchar(10)    | NO   | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

QUERY:

```
-----
DESCRIBE WorksOn;
```

OUTPUT:

```
-----
+-----+-----+-----+-----+-----+-----+
| Field       | Type           | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| empNo       | varchar(10)    | NO   | PRI | NULL    |       |
| projNo      | varchar(10)    | NO   | PRI | NULL    |       |
| dateWorked  | date           | NO   | PRI | NULL    |       |
| hoursWorked | int(11)        | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
=====
INSERT SAMPLE DATA
=====
```

```
-----
Insert Employees (no deptNo yet due to FK)
-----
```

QUERY:

```
-----
INSERT INTO Employee VALUES
('E1', 'John', 'Smith', '123 Main St', '1990-01-01', 'M', 'Manager',
NULL),
('E2', 'Sarah', 'Jones', '456 High St', '1985-05-15', 'F', 'Analyst',
NULL),
('E3', 'Mike', 'Brown', '789 Park Rd', '1992-03-20', 'M', 'Software Developer',
NULL),
('E4', 'Lisa', 'Wilson', '321 Lake Ave', '1988-07-10', 'F', 'Manager',
NULL),
('E5', 'James', 'Taylor', '654 Oak St', '1995-11-25', 'M', 'Team Leader',
NULL);
```

OUTPUT:

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

Insert Departments

QUERY:

INSERT INTO Department VALUES

('D1', 'IT', 'E1'),

('D2', 'Finance', 'E2'),

('D3', 'Marketing', 'E4');

OUTPUT:

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

Update Employee deptNo (now Department exists)

QUERY:

UPDATE Employee SET deptNo = 'D1' WHERE empNo = 'E1';

UPDATE Employee SET deptNo = 'D2' WHERE empNo = 'E2';

UPDATE Employee SET deptNo = 'D1' WHERE empNo = 'E3';

UPDATE Employee SET deptNo = 'D3' WHERE empNo = 'E4';

UPDATE Employee SET deptNo = 'D2' WHERE empNo = 'E5';

OUTPUT:

Query OK, 1 row affected (0.01 sec)

Query OK, 1 row affected (0.01 sec)

Query OK, 1 row affected (0.01 sec)

Query OK, 1 row affected (0.01 sec)

Query OK, 1 row affected (0.01 sec)

Insert Projects

QUERY:

```
-----
INSERT INTO Project VALUES
('P1', 'SCCS', 'D1'),
('P2', 'Payroll', 'D2'),
('P3', 'Marketing', 'D3');
```

OUTPUT:

```
-----
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
-----
Insert WorksOn Records
-----
```

QUERY:

```
-----
INSERT INTO worksOn VALUES
('E1', 'P1', '2024-01-01', 8),
('E1', 'P2', '2024-01-02', 6),
('E2', 'P1', '2024-01-01', 7),
('E3', 'P1', '2024-01-03', 5),
('E1', 'P1', '2024-01-04', 4),
('E4', 'P3', '2024-01-01', 8),
('E5', 'P2', '2024-01-02', 6),
('E2', 'P2', '2024-01-03', 5);
```

OUTPUT:

```
-----
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

```
-----
Verify Inserted Data – SELECT from each table
-----
```

QUERY: SELECT * FROM employee;

OUTPUT:

```
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
| empNo | fName | lName | address      | DOB       | sex | position          |
deptNo |
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
| E1    | John  | Smith | 123 Main St  | 1990-01-01 | M   | Manager           |
```

```

D1      |
| E2    | Sarah | Jones | 456 High St | 1985-05-15 | F   | Analyst          |
D2      |
| E3    | Mike  | Brown | 789 Park Rd | 1992-03-20 | M   | Software Developer |
D1      |
| E4    | Lisa  | Wilson | 321 Lake Ave | 1988-07-10 | F   | Manager          |
D3      |
| E5    | James | Taylor | 654 Oak St  | 1995-11-25 | M   | Team Leader      |
D2      |
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+
5 rows in set (0.126 sec)

```

QUERY:

SELECT * FROM Department;

OUTPUT:

```

+-----+-----+-----+
| deptNo | deptName | mgrEmpNo |
+-----+-----+-----+
| D1     | IT       | E1       |
| D2     | Finance  | E2       |
| D3     | Marketing| E4       |
+-----+-----+-----+
3 rows in set (0.00 sec)

```

QUERY:

SELECT * FROM Project;

OUTPUT:

```

+-----+-----+-----+
| projNo | projName | deptNo |
+-----+-----+-----+
| P1     | SCCS     | D1     |
| P2     | Payroll  | D2     |
| P3     | Marketing| D3     |
+-----+-----+-----+
3 rows in set (0.00 sec)

```

QUERY:

SELECT * FROM WorksOn;

OUTPUT:

```
-----
+-----+-----+-----+-----+
| empNo | projNo | dateWorked | hoursWorked |
+-----+-----+-----+-----+
| E1     | P1     | 2024-01-01 | 8           |
| E1     | P2     | 2024-01-02 | 6           |
| E2     | P1     | 2024-01-01 | 7           |
| E3     | P1     | 2024-01-03 | 5           |
| E1     | P1     | 2024-01-04 | 4           |
| E4     | P3     | 2024-01-01 | 8           |
| E5     | P2     | 2024-01-02 | 6           |
| E2     | P2     | 2024-01-03 | 5           |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

```
=====
TESTING CONSTRAINTS (7.24 a, b, c)
=====
```

```
-----
(a) Test sex constraint – should FAIL
-----
```

QUERY:

```
-----
INSERT INTO Employee VALUES
('E6', 'Tom', 'Gray', '999 Test St', '2000-01-01', 'X', 'Analyst', NULL);
```

OUTPUT:

```
-----
ERROR 4025 (23000): CONSTRAINT `chk_sex` failed for `ProjectsDB`.`Employee`
```

```
-----
(b) Test position constraint – should FAIL
-----
```

QUERY:

```
-----
INSERT INTO Employee VALUES
('E6', 'Tom', 'Gray', '999 Test St', '2000-01-01', 'M', 'Director', NULL);
```

OUTPUT:

```
-----
ERROR 4025 (23000): CONSTRAINT `chk_position` failed for `ProjectsDB`.`Employee`
```

(c) Test hoursWorked constraint – should FAIL

QUERY:

INSERT INTO WorksOn VALUES ('E1', 'P1', '2024-02-01', 50);

OUTPUT:

ERROR 4025 (23000): CONSTRAINT `chk\_hours` failed for `ProjectsDB`.`WorksOn`

=====

7.25 – VIEW: Projects Managed by Female Managers

=====

QUERY:

CREATE VIEW FemaleManagerProjects AS
 SELECT p.projNo, p.projName, p.deptNo
 FROM Project p
 JOIN Department d ON p.deptNo = d.deptNo
 JOIN Employee e ON d.mgrEmpNo = e.empNo
 WHERE e.sex = 'F'
 ORDER BY p.projNo;

OUTPUT:

Query OK, 0 rows affected (0.01 sec)

QUERY:

SELECT * FROM FemaleManagerProjects;

OUTPUT:

+-----+-----+-----+
| projNo | projName | deptNo |
+-----+-----+-----+
| P2 | Payroll | D2 |
| P3 | Marketing | D3 |
+-----+-----+-----+
2 rows in set (0.00 sec)

NOTE: P2 managed by Sarah Jones (F), P3 managed by Lisa Wilson (F)

=====

7.26 – VIEW: EmpProjectDetails

=====

QUERY:

```
CREATE VIEW EmpProjectDetails AS
  SELECT e.empNo, e.fName, e.lName, p.projName, w.hoursWorked
  FROM   Employee e
  JOIN   WorksOn  w ON e.empNo = w.empNo
  JOIN   Project  p ON w.projNo = p.projNo;
```

OUTPUT:

Query OK, 0 rows affected (0.01 sec)

QUERY:

```
SELECT * FROM EmpProjectDetails;
```

OUTPUT:

empNo	fName	lName	projName	hoursWorked
E1	John	Smith	SCCS	8
E1	John	Smith	Payroll	6
E2	Sarah	Jones	SCCS	7
E3	Mike	Brown	SCCS	5
E1	John	Smith	SCCS	4
E4	Lisa	Wilson	Marketing	8
E5	James	Taylor	Payroll	6
E2	Sarah	Jones	Payroll	5

8 rows in set (0.00 sec)

=====

7.27 – CREATE EmpProject VIEW

=====

QUERY:

```
CREATE VIEW EmpProject(empNo, projNo, totalHours)
AS SELECT w.empNo, w.projNo, SUM(hoursWorked)
  FROM   Employee e, Project p, WorksOn w
  WHERE  e.empNo = w.empNo AND p.projNo = w.projNo
  GROUP BY w.empNo, w.projNo;
```

OUTPUT:

Query OK, 0 rows affected (0.01 sec)

7.27 (a) – SELECT * FROM EmpProject

QUERY:

SELECT * FROM EmpProject;

OUTPUT:

+-----+-----+-----+
| empNo | projNo | totalHours |
+-----+-----+-----+
E1	P1	12
E1	P2	6
E2	P1	7
E2	P2	5
E3	P1	5
E4	P3	8
E5	P2	6
+-----+-----+-----+

7 rows in set (0.00 sec)

NOTE: Shows total hours each employee worked on each project

7.27 (b) – SELECT projNo WHERE projNo = 'SCCS'

QUERY:

SELECT projNo
FROM EmpProject
WHERE projNo = 'P1';

OUTPUT:

+-----+
| projNo |
+-----+
| P1 |
| P1 |
| P1 |
+-----+

3 rows in set (0.00 sec)

NOTE: 3 rows because E1, E2, and E3 all worked on project P1 (SCCS)

7.27 (c) – COUNT how many projects E1 worked on

QUERY:

SELECT COUNT(projNo)
FROM EmpProject
WHERE empNo = 'E1';

OUTPUT:

+-----+
| COUNT(projNo) |
+-----+
| 2 |
+-----+
1 row in set (0.00 sec)

NOTE: E1 (John Smith) worked on 2 projects – P1 and P2

7.27 (d) – Grand Total Hours per Employee

QUERY:

SELECT empNo, SUM(totalHours) AS grandTotalHours
FROM EmpProject
GROUP BY empNo;

OUTPUT:

+-----+-----+
| empNo | grandTotalHours |
+-----+-----+
E1	18
E2	12
E3	5
E4	8
E5	6
+-----+-----+
5 rows in set (0.00 sec)

NOTE: E1 total = 12(P1) + 6(P2) = 18
E2 total = 7(P1) + 5(P2) = 12

```
=====
CREATE TABLE PART AND INSERT VALUES
=====
```

QUERY:

```
CREATE TABLE Part (
->     partNo  VARCHAR(10)    NOT NULL,
->     contract VARCHAR(10)    NOT NULL,
->     partCost DECIMAL(10,2) NOT NULL,
->
->     CONSTRAINT pk_part PRIMARY KEY (partNo, contract)
-> );
```

OUTPUT:

Query OK, 0 rows affected, 1 warning (0.035 sec)

QUERY:

```
INSERT INTO Part VALUES
-> ('P001', 'C1', 1200.00),
-> ('P001', 'C2', 800.00),
-> ('P002', 'C1', 500.00),
-> ('P002', 'C3', 1500.00),
-> ('P003', 'C2', 300.00),
-> ('P003', 'C4', 250.00),
-> ('P004', 'C1', 2000.00),
-> ('P005', 'C3', 750.00);
```

OUTPUT:

Query OK, 8 rows affected (0.032 sec)

Records: 8 Duplicates: 0 Warnings: 0

```
=====
CREATE EXPENSIVEPART VIEW
=====
```

QUERY:

```
CREATE View ExpensiveParts(PartNo) AS SELECT DISTINCT partNo FROM Part WHERE
partCost > 1000;
```

OUTPUT:

Query OK, 0 rows affected (0.034 sec)

```
=====
VERIFY ALL VIEWS
=====
```

```
MariaDB [projects]> SHOW FULL TABLES WHERE Table_type = 'VIEW';
```

Tables_in_projects	Table_type
empproject	VIEW
empprojectdetails	VIEW
expensiveparts	VIEW
femalemanagerprojects	VIEW

```
4 rows in set (0.030 sec) SHOW FULL TABLES WHERE Table_type = 'VIEW';
```